

Kumar Arpit

3rd Year Undergraduate
Department of Computer Science and Engineering

✉ arpit20@iitk.ac.in ☎ +91-9430000099
in LinkedIn | GitHub

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2020 - Present	B.Tech	Indian Institute of Technology Kanpur	8.8/10
2020	Class XII (CBSE)	Dr. GL. Dutta DAV Public School, Patna	93.8%
2018	Class X (ICSE)	Don Bosco Academy, Patna	96.4%

Scholastic Achievements

- Awarded **Academic Excellence Award** by IIT Kanpur for exceptional academic performance in the year 2020.
- Secured **All India Rank 707** in **JEE Advanced 2020** amongst the 1.6 lakh shortlisted candidates.
- Secured **All India Rank 3521** in **JEE Mains 2020** amongst the 10.3 lakh candidates.

Key Projects

MCMC Methods in Julia | Stamatics, IITK  Apr 2022–Jun 2022

- Implemented **Pseudo-random generators** and discrete **inverse transform** and discrete accept reject samplers.
- Used continuous **Accept-reject** to sample from p-dimensional **hypersphere** and a **truncated normal** distributions.
- Looked into and implemented a weighted **importance sampling estimator**. Looked into **Bayesian models**.
- Implemented the MCMC **Metropolis-Hastings** algorithm to sample from several complex posterior distributions.

Blocks and Chains | Association for Computing Activities IITK  Apr 2021–Jul 2021

- Used **crypto** library in **JavaScript** for **hashing**, **key-pair generation** and **digitally signing** documents.
- Developed a program to generate **SHA256** hash for any input file by implementing the algorithm from scratch.
- Implemented functions to **mine** a block in a chain and to **generate and verify** the **transaction data** for the same.

Demystifying Algorithms | Programming Club, IITK Apr 2022–Ongoing

- Used segment trees and **Fenwick trees** data structures and **sqrt decomposition** to implement range query algorithms.
- Solved CP questions using prime number sieves, **bitmask dp**, **meet in the middle** and other number theoretic algorithms.

Modern Incentives In Computer Science | Science Coffee House, IITK May 2021–Oct 2021

- Looked into **mechanism design** and **DSIC Auctions** for Surplus maximisation and Revenue maximization problems.
- Analysed the matching problem in one and **two sided markets**, fair division and **strategyproofness** in **voting**.
- Analysed **prisoner's dilemma**, strict dominance and **Nash and market equilibria** in Game theory.

Course Projects

IITKart | CS253 | Dr. Indranil Saha  Jan 2022–Apr 2022

- Collaborated with a **team** of ten members to deploy a buying-selling portal for the campus based on **layered architecture**.
- Implemented the backend for the website using **Django** framework with **SQLite** as database, and hosted it on **Heroku**.
- Contributed to Software Requirement Specification, Software Design, Implementation, Testing and User Manual documents.

SAT Solver | CS202 | Dr. Subhajit Roy  Jan 2022–Feb 2022

- Generated a pseudo-random **maximal Sudoku** and efficiently encoded it as **short CNF Clauses** to solve using **pySAT**.
- Implemented a SAT solver for CNF formulae using the **DPLL** algorithm by optimising backtracking algorithm with **greedy heuristics** such as minimal literal choice, **Unit Propagation** and Pure Literal Elimination.

2-3 Trees and Polynomial Algebra in C | ESO207 | Dr. Anil Seth  Aug 2021–Oct 2021

- Implemented the data structure for **self balancing 2-3 tree** in the C Language and algorithms to create and **merge** them.
- Implemented a function from scratch in C to add and multiply two **arbitrarily large polynomials** using **Linked lists**.

Technical Skills * Exposure

- **Languages:** C, C++, Python, JavaScript, MIPS Assembly, JAVA*, Rust*, Julia*
- **Utilities:** Git, Linux, Numpy, Latex, Bash, Django, TensorFlow, HTML, Verilog, MicroCap

Relevant Coursework * Ongoing Courses

Data Structure & Algorithms (ESO207)	Software Development and Operations (CS253)	Probability and Statistics (CS203)
Fundamentals of Computing (ESC101)	Real Analysis and Calculus (MTH101)	Logic in CS (CS202)
Computer Organization (CS220)	Linear Algebra and ODE (MTH102)	Discrete Mathematics (CS201)

Positions of Responsibility

Coordinator | Anime Club, IIT Kanpur Apr-2022 - Present

- Leading a team of over 20 secretaries to manage events like quizzes and showdowns and content like AMVs and artworks.
- Enhanced the reach of the club on various social media platforms, Instagram, Facebook and Twitter by over **160%**.

Project Mentor - Reinforcement Learning | Association for Computing Activities IITK  Jun-2022 - Present

- Co-mentoring over 15 students from CSE Department in learning **Reinforcement Learning** from scratch.
- Conducted weekly assignments, published tutorial blogs and had in-person sessions for discussions and doubts.

Academic Mentor - CHM102 | Counselling Service, IIT Kanpur Oct-2021 - Jul-2022

- Conducted lectures on general chemistry at **institute level** to clarify doubts of students before their semester exams.
- Mentored several weak students by designing assignments, solving doubts and taking one to one sessions with them.